Residual collector: one addition per step, and distinguished points

SUPERSEDED (w > 0 figures). The per-step distinguished-point test of `residual-fused-dp` was not in the priced counters when this study ran, so the w = 0.1 and w = 1 results and the distances to rho below understate priced work. The w = 0 figures are unaffected. Corrected, priced rerun: `research/ic_residual_dp_eq_20261101`.

What was implemented. Two target modes in `src/cryptanalysis/ic_framework/residual.rs`, both leaving the baseline path (`targets=residual`) unchanged: `residual-fused` takes one group addition per walk step from a precomputed table of $J_j - P_i$ (16·|F| setup additions, counted); `residual-fused-dp` adds distinguished points: one DP test per step (counted native op, the analogue of rho's per-step canonicalisation), and only residuals in a 2^{-b} fraction ($b = \lfloor \log_2 r / 2 \rfloor - 9$; 0 for $\log_2 r < 20$, 8 for the 36–37-bit curves) are tested against F, probed and stored. Counts are in ecbench's counted unit; every answer verified by the harness; `ecbench verify --replay 12` audits OK (2048/2048 records per study).

Result 1 (study 1, `research/ic_residual_fused_20261031`): fused step

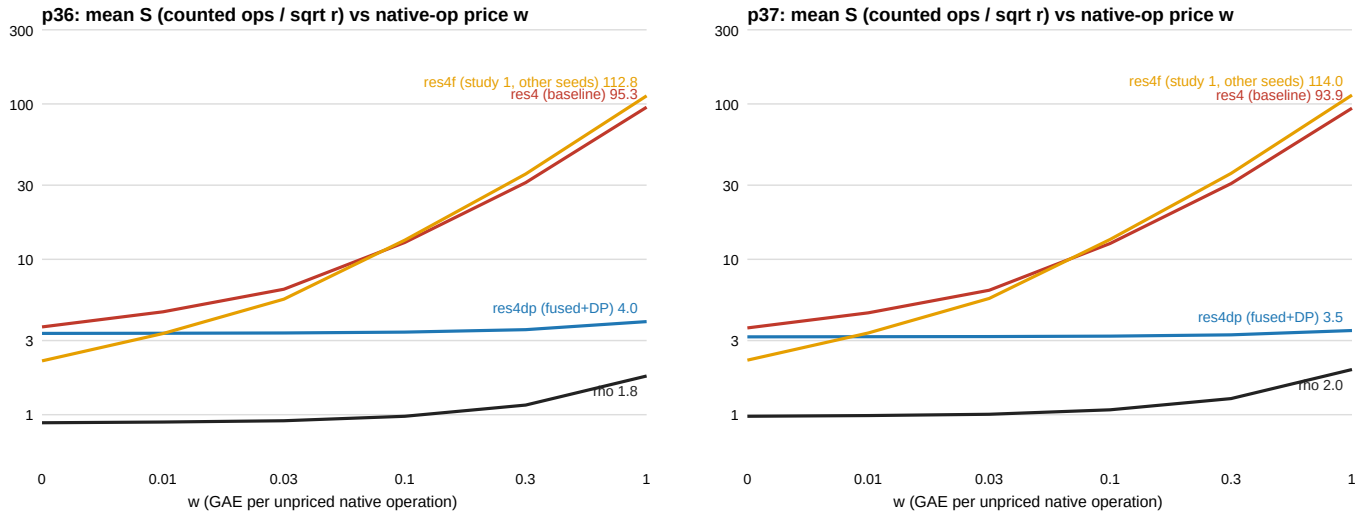
gae/step falls from 2.005 to 1.004. Fused/baseline counted S at w = 0: 0.56–0.92 (all intervals below 1, 12 cells); at w = 0.1: 0.91–1.13; at w = 1: 1.06–1.30. The preregistered rule (below 1 at w = 0, 0.1 and 1) is **not met**: the saving is real at w = 0 but the priced table work (about 2 lookups and 56 stored bytes per step) dominates at w ≥ 0.1.

Result 2 (study 2, `research/ic_residual_dp_20261101`): fused + distinguished points, paired by instance, 16 targets × 3 rounds

S(candidate)/S(baseline), 95% interval; green = interval below 1. p36, p37 are the DP-active curves (the claim); p16, p18 have b = 0, so the candidate equals the fused walk there (control).

curve	pair	w = 0	w = 0.1	w = 1
p36	res2dp / res2	0.875 [0.747, 1.026]	0.329 [0.281, 0.386]	0.055 [0.047, 0.065]
p36	res4dp / res4	0.908 [0.852, 0.967]	0.265 [0.249, 0.282]	0.042 [0.039, 0.044]
p36	res8dp / res8	0.893 [0.837, 0.953]	0.181 [0.170, 0.193]	0.028 [0.026, 0.030]
p37	res2dp / res2	0.799 [0.720, 0.887]	0.298 [0.269, 0.331]	0.047 [0.043, 0.053]
p37	res4dp / res4	0.876 [0.799, 0.960]	0.253 [0.231, 0.278]	0.037 [0.034, 0.041]
p37	res8dp / res8	0.853 [0.784, 0.928]	0.170 [0.156, 0.185]	0.024 [0.022, 0.026]
p16	res2dp / res2	0.881 [0.850, 0.914]	1.039 [0.973, 1.110]	1.208 [1.103, 1.324]
p16	res4dp / res4	0.863 [0.821, 0.907]	1.004 [0.921, 1.095]	1.103 [0.988, 1.231]
p16	res8dp / res8	0.901 [0.871, 0.933]	1.070 [0.993, 1.153]	1.140 [1.043, 1.247]
p18	res2dp / res2	0.779 [0.732, 0.829]	0.970 [0.885, 1.064]	1.126 [1.006, 1.260]
p18	res4dp / res4	0.786 [0.744, 0.830]	1.006 [0.925, 1.094]	1.127 [1.023, 1.242]
p18	res8dp / res8	0.851 [0.812, 0.892]	1.165 [1.079, 1.258]	1.270 [1.167, 1.383]

Frozen rule C1 (all six DP-active cells below 1 at w = 0, 0.1, 1): **5 of 6 cells hold; C1 is not met strictly** — p36 res2 at w = 0 has interval [0.747, 1.026]. At w = 0.1 and w = 1 all six cells are far below 1 (0.17–0.33 and 0.024–0.055). A/A (res4 vs res4aa) = 1.0000 on every curve.



Mean S per arm vs w (study-2 sessions for rho, res4, res4dp; res4f from study 1, different seeds, shown for context only).

Distance remaining to matched rho (S ratio rho → arm; 1.00 = parity)

curve	arm	w = 0	w = 0.1	w = 1
p36	res4	4.14	13.14	53.70
p36	res4dp	3.76	3.48	2.24
p37	res4	3.70	11.76	48.06
p37	res4dp	3.24	2.98	1.78

The order holds across the band: rho is ahead at every weight. Baseline res4 was 4.1 / 13.1 / 53.7× (p36) and 3.7 / 11.8 / 48.1× (p37); res4dp is 3.76 / 3.48 / 2.24× and 3.24 / 2.98 / 1.78×. The w = 1 distance fell about 24–27×; the w = 0 distance fell about 9–12%.

Classification and caveats

- Class: **engineering** improvement of a generic collision search (lower priced native work: lookups and stored bytes), not a structural index-calculus advance; the curve-free-null conclusion of `ic_prime_residual_lp_20261026` carries over. It lowers the pipeline's counted cost against the frozen baseline on the same instances.
- The candidate does not merge L with −L after a collision, so it needs about 1.5× more steps than the fused no-DP walk (exploratory, 16–24 targets). A negation-equivariant step was tried and lost to short cycles (walks trapped until the cap); not part of either study.
- Run locally at isolation L0 (isolab-eks route unavailable here; counts are deterministic and carry no wall-clock claim). L2 specs are committed for a later isolab-eks replay.
- Only 4 curves, two DP-active, both ≈2^36; behaviour at larger r is not measured. No validation-gate artifacts (replay certificate, validator report, §5a-banded matched-rho `VALIDATION.json`) exist yet, so no leaderboard or `benchmarks/README.md` row is added.

Sources

Teske, *On random walks for Pollard's rho method*, Math. Comp. 70 (2001) 809–825 (r-adding walks); Bos–Kleinjung–Lenstra, *On the use of the negation map in the Pollard rho method*, ANTS-IX, LNCS 6197 (2010) 66–82 (fruitless cycles; verified by search earlier in this session); van Oorschot–Wiener, *Parallel collision search with cryptanalytic applications*, J. Cryptology 12 (1999) 1–28 (distinguished points; cited from memory, not re-read). Repository: `research/ic_prime_residual_lp_20261026`, `research/ic_planner_exhaustive_20261026`.